ModelIt!

Teacher's Guide

Why Do Magnets Work Through Your Phone Case?

Modeling Electromagnetic Fields, Force at a Distance, and Energy Storage

Quick Reference

Grade Level	Grade 10L1
NGSS Standard	HS-PS2-3, HS-PS3-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS2-3, HS-PS3-5

Apply scientific and engineering ideas to design, evaluate, and refine a device that minimizes the force on a macroscopic object during a collision. Develop and use a model of two objects interacting through electric or magnetic fields to illustrate the forces between objects and the changes in energy of the objects due to the interaction.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop and use models of magnetic field interactions to predict forces between objects at various distances and through different materials.
Disciplinary Core Idea	**PS2.B Types of Interactions / PS3.C Relationship Between Energy and Forces** — When two objects interact through magnetic fields, the force depends on their magnetic properties, distance, and orientation. Energy is stored in the field and converts during interactions.
Crosscutting Concept	**Systems and System Models** — Students model the magnet-field-material system to understand how invisible fields mediate forces between objects and store energy that can be transferred during interactions.

CAST Testing Alignment

MULTIPLE CHOICE:

A neodymium magnet holds a metal plate firmly through a 3mm plastic phone case. If the case is replaced with a 3mm iron plate, what will happen to the force on the metal plate behind it, and why?

CONSTRUCTED RESPONSE:

Using your model, explain why a magnet can attract a paperclip through a wooden table but the force drops dramatically when you increase the distance by just a few centimeters. Include the roles of field permeation through materials, the inverse-distance relationship, and field energy in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Magnets are one of the most familiar yet least understood phenomena in everyday life. The fact that a magnet can exert force on metal through solid materials — without any physical connection — seems almost magical. But it reveals something profound about the nature of reality: space is not empty. It is filled with fields — invisible physical entities that carry energy, exert forces, and mediate every electromagnetic interaction in the universe.

What Is a Magnetic Field?

A magnetic field is a physical entity that exists in the space around any magnet or moving electric charge. It's not just a mathematical tool — it contains real energy, can exert real forces, and propagates through space at the speed of light. Field lines represent the direction a compass needle would point at each location. The field is strongest where lines are densest (near the poles) and weakest where they're spread apart (far from the magnet).

Fields and Materials

Magnetic fields interact differently with different materials. Ferromagnetic materials (iron, nickel, cobalt) have atomic magnetic domains that align with the field, concentrating and redirecting it — this is why iron shields and redirects magnetic fields. Diamagnetic and paramagnetic materials (plastic, wood, glass, water, air, aluminum) have virtually no interaction with the field — it passes through them as if they weren't there. This is why your phone case is invisible to a magnet.

The Distance Relationship

For a magnetic dipole (bar magnet), the field strength decreases approximately as $1/r^3$ — the inverse CUBE of distance. This is even steeper than the inverse-square law of gravity. Doubling the distance reduces the force to about 1/8 (not 1/4). This explains why magnets seem to suddenly 'grab' when close — the force increases explosively as distance shrinks. At 1mm, a neodymium magnet might exert 5N; at 10mm, only about 0.005N — a thousand-fold decrease.

Energy in the Field

The magnetic field stores energy — about $B^2/(2\mu_0)$ joules per cubic meter. When a magnet attracts a paperclip, the total field energy decreases as the paperclip moves closer (a lower-energy configuration). This decrease in field energy converts to the kinetic energy of the accelerating paperclip. This is not abstract — the energy in the field is as real as the energy in a stretched spring. In fact, every electric motor, generator, and transformer works by converting energy between magnetic fields and mechanical motion.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify magnetic field strength, distance from magnet, material type, force of attraction, and energy stored in field as the key components of the magnetic interaction system.

E **Establish Relationships**

Students determine that field strength sets the maximum possible force, distance dramatically reduces force following the inverse cube relationship, and material type determines whether the field passes through unimpeded or is shielded.

V **Visualize & Model**

Students build a computational model showing how magnetic force varies with distance and material, and how field energy converts to kinetic energy during attraction.

E **Evaluate Outcomes**

Students run direct contact, through-phone-case, and distance falloff scenarios to discover the inverse cube relationship and the transparency of non-magnetic materials to fields.

R **Revise & Extend**

Students add magnetic domain alignment, temperature effects, or electromagnetic induction to model more complete electromagnetic behavior.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Your phone sticks to a magnetic mount through a plastic case. The magnet never touches the metal. How does force travel through solid material?"*

Watch For

Pass around a strong magnet and various materials. Students test: Does the magnet work through paper? Wood? Plastic? Aluminum? Steel?

Slide 2: Learning Objectives

3 min

Say: *"Today we're investigating one of the most fundamental questions in physics: how can objects push and pull each other without touching?"*

Watch For

Have students read objectives. Pre-teach 'magnetic field' and 'inverse square law' with visual demonstrations.

Slide 3: Big Question

3 min

Say: *"A magnet on the outside of your phone case pulls on metal inside the case. What is in the space between them that transmits this force?"*

Watch For

Students draw what they think exists between a magnet and a paperclip. Most will draw 'nothing' — challenge this assumption.

Slide 4: LEVER Framework

2 min

Say: *"We're modeling the invisible — building a computational model of something you can't see but can definitely feel."*

Watch For

Preview LEVER steps. Emphasize we're exploring FIELDS — real physical entities that fill 'empty' space.

Slide 5: Activity 1: Components

8 min

Say: *"What determines how strongly a magnet pulls? Which factors can you change in a design?"*

Watch For

Guide sorting. Field strength, distance, and material are design choices; force and energy are physics outcomes.

Slide 6: Activity 2: Connections

8 min

Say: *"Double the distance and the force drops to one-eighth. But a plastic case? Almost no effect. Why the difference?"*

Watch For

Build the relationship map. Introduce the inverse cube law with numerical examples. Test materials.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's measure exactly how force changes with distance — and prove that your phone case is invisible to the magnetic field."*

Watch For

Students predict force at various distances. Run simulations. Compare through air, plastic, glass, and iron.

Slide 8: Discoveries

5 min

Say: *"The space between a magnet and metal is NOT empty. It's filled with a field that carries real energy. When the paperclip flies to the magnet, that's field energy converting to motion."*

Watch For

Show iron filing field visualization. Discuss: The field is real. It carries energy. It's everywhere.

Slide 9: STEM Challenge

12 min

Say: *"Design a phone mount that holds securely through any case, survives bumpy roads, but comes off with one hand. Use your model data."*

Watch For

Groups specify magnet types, sizes, arrangements, and metal plate requirements. Test with force calculations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Magnetic Field Strength, Distance from Magnet, Material Type
- **INTERNAL:** Force of Attraction, Energy Stored in Field

Relationships

- **Magnetic Field Strength to Force of Attraction:** POSITIVE (+) — Stronger magnetic fields exert greater forces on ferromagnetic objects. A neodymium magnet (field ~1.2 Tesla) produces roughly 10-100 times more force than a ceramic magnet (~0.1 Tesla) at the same distance.
- **Distance from Magnet to Force of Attraction:** NEGATIVE (inverse cube) — Force decreases approximately as the cube of distance for magnetic dipoles: $F \propto 1/r^3$. Doubling distance reduces force to about 1/8. This extremely steep falloff explains why magnets seem to 'grab' suddenly when close — the force increases explosively as distance shrinks.
- **Material Type to Force of Attraction:** CONDITIONAL — Non-ferromagnetic materials (plastic, glass, wood, air) have negligible effect on the magnetic field — it passes through them transparently. Ferromagnetic materials (iron, steel) redirect and shield the field, potentially reducing the force on objects behind them. Material type acts as a gate: either transparent or opaque to the field.

Key Discoveries

- Magnetic fields pass through non-ferromagnetic materials (plastic, glass, wood, air, water) with virtually no reduction — the phone case is invisible to the magnetic field
- Force decreases with the cube of distance for magnetic dipoles — doubling the distance reduces force to about one-eighth, making the falloff extremely steep
- The magnetic field contains real energy that can do real work — when a magnet pulls a paperclip, the field's stored energy converts to the paperclip's kinetic energy
- Fields are not just mathematical descriptions — they are physical entities that carry energy, exert forces, and travel through space at the speed of light

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Magnets push or pull through 'magic' with no mechanism"	Magnetic force is mediated by the magnetic field — a real, physical entity that fills the space around every magnet. The field contains energy, exerts forces, and obeys precise mathematical laws. There is nothing mysterious about it: the field is produced by moving charges (electron orbital motion and spin) and interacts with other magnetic materials through the alignment of atomic magnetic moments. Maxwell's equations describe this completely.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Magnetic force goes through everything equally"	While magnetic fields pass through non-ferromagnetic materials (plastic, glass, air) with virtually no reduction, ferromagnetic materials (iron, steel) strongly interact with and redirect the field. An iron plate between a magnet and target will shield the target by 'absorbing' the field into itself. This is why steel is used for magnetic shielding. The field doesn't go through everything equally — it depends entirely on the material's magnetic properties.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Magnets eventually 'run out' of force or energy"	Permanent magnets don't 'run out' of magnetism through normal use. The magnetic field is maintained by the quantum mechanical spin of electrons in the magnet's crystal structure — this is an intrinsic property of the material, not a consumable resource. However, magnets CAN be demagnetized by excessive heat (above the Curie temperature), physical shock (disrupts domain alignment), or opposing magnetic fields. But under normal conditions, a neodymium magnet loses less than 1% of its field strength per century.